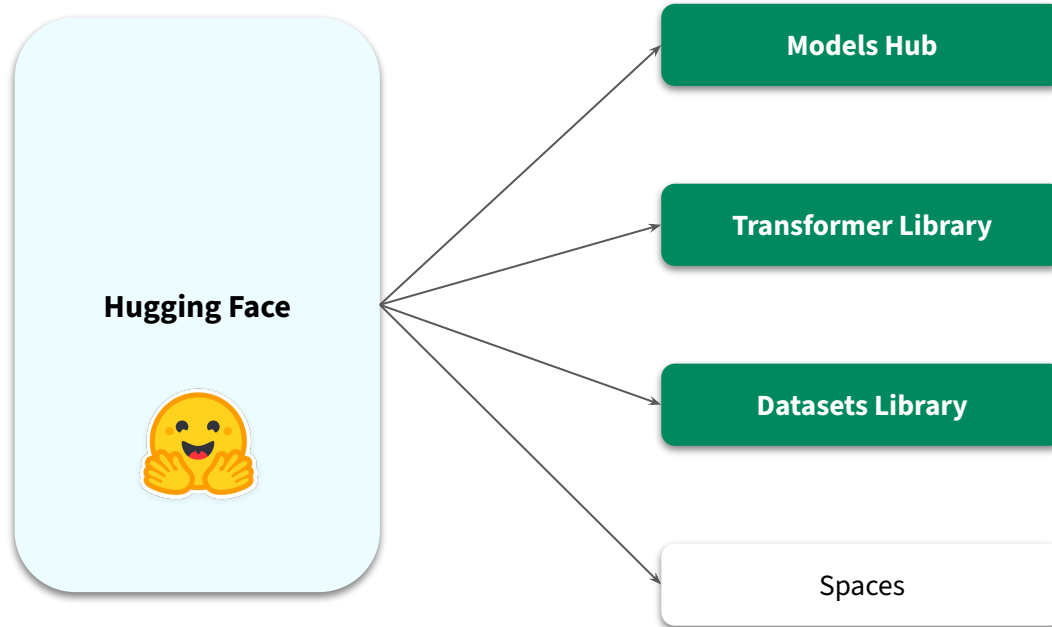


What is Hugging Face?

Hugging Face Basics

Key Offerings



Hugging Face Models Hub

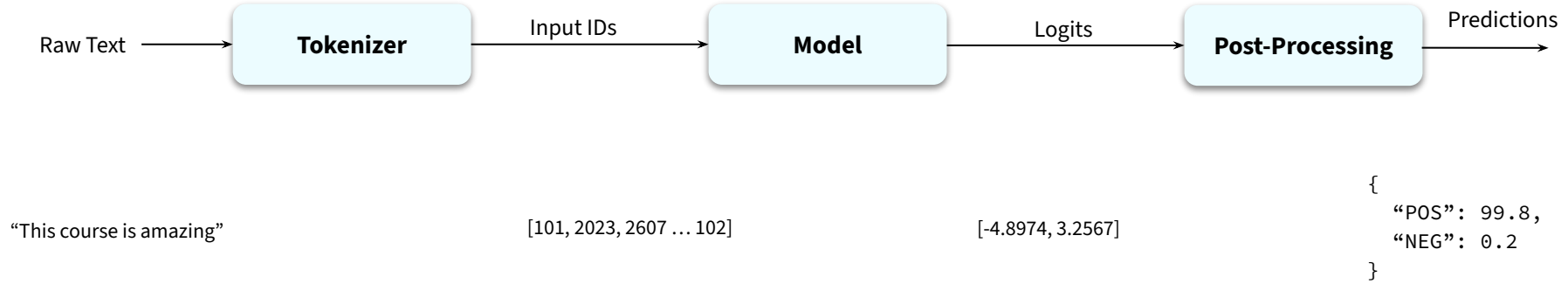
Hugging Face Basics

The Pipeline Abstraction

Hugging Face Basics

Pipeline abstracts away the complexity





Variations of Transformer

Hugging Face Basics



Why do we even need variations of Transformers?

The Reader

Encoder-Only Models

Sometimes you don't need to generate output text. Sometimes, you just need to predict something on the input text.

Text classification, Sentiment Analysis, NER, etc.

The Writer

Decoder-Only Models

Sometimes you don't a lot of input information to generate output text. Sometimes, you just need to output stuff.

Next-word prediction for Chatbots

The Translator

Encoder-Decoder Models

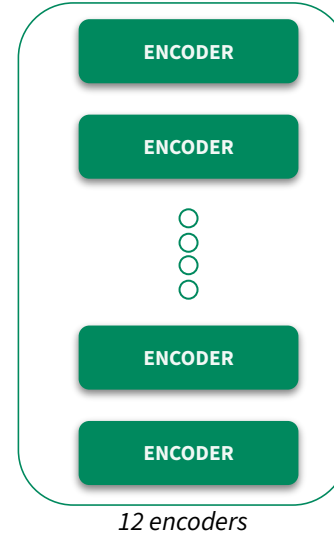
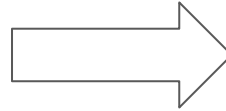
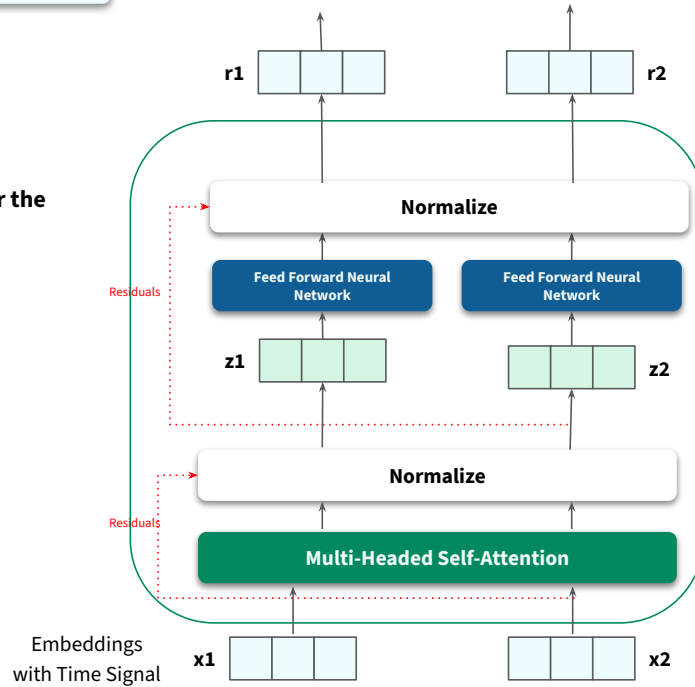
These are your standard transformer models that we have already looked at.

Translation mechanisms

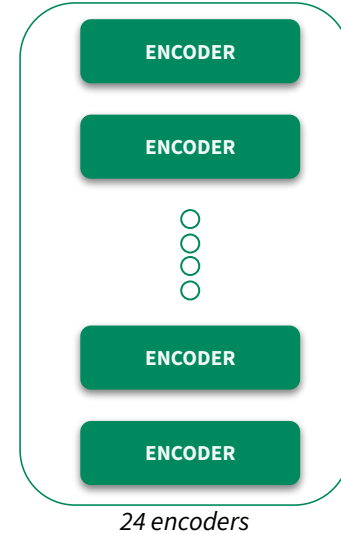
BERT: BiDirectional Encoder Representations from Transformers

Encoder-Only

Remember the encoder?



BERT_{base}



BERT_{large}



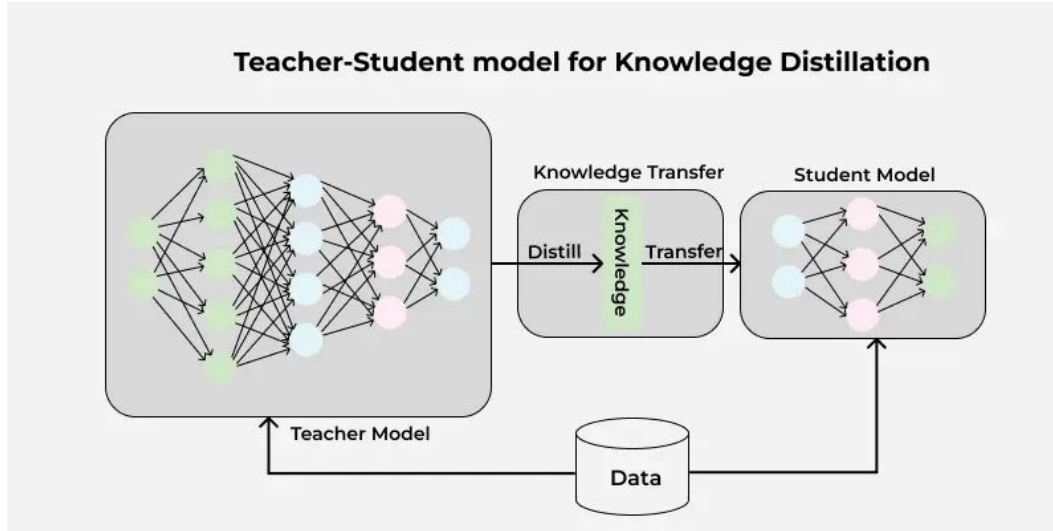
BERT: BiDirectional Encoder Representations from Transformers

Encoder-Only

- Keeps only the encoder half.
- Reads the entire input sentence in **both directions**: left-to-right and right-to-left to better understand the sentence.
- Trained using fill-in-the-blank training: hides random words in a sentence and forces the model to guess them based on surrounding context. Also, called **Masked Language Modeling**.
- Performs best on fill-mask, sentiment analysis and text classification tasks.

DistilBERT

Encoder-Only



- A miniature version of BERT with half the layers, making it much smaller and faster.
- Teacher-Student Learning: It copies a fully trained BERT model, learning to mimic its behavior while being 40% smaller and 60% faster, whilst retaining 97% of the original performance. Also, known as **Knowledge Distillation**.
- Performs best on fill-mask, sentiment analysis and text classification tasks.



RoBERTa: Robustly Optimized BERT Pretraining Approach

Encoder-Only

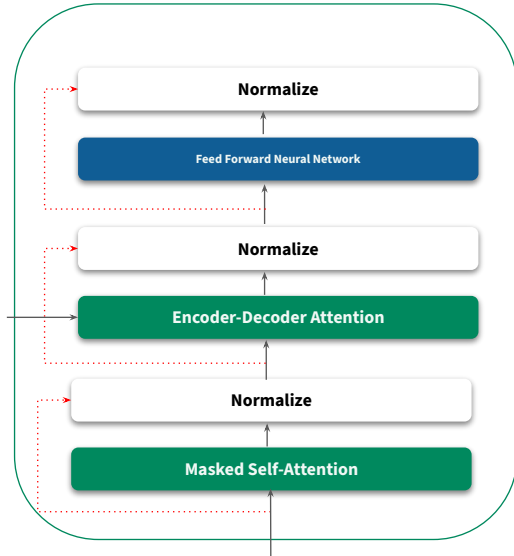
- Same exact structure as BERT, but trained with way more data, bigger batches, and for a much longer time.
- **Dynamic Masking:** Instead of hiding the same words in a sentence every time, it changes which words are hidden in every training round.
- Performs best on fill-mask, sentiment analysis and text classification tasks.



GPT2: Generative Pre-trained Transformer 2

Decoder-Only

Remember the decoder?



12 DECODERS

GPT2_{base}

24 DECODERS

GPT2_{medium}

36 DECODERS

GPT2_{large}

48 DECODERS

GPT2_{XL}



GPT2: Generative Pre-trained Transformer 2

Decoder-Only

- Keeps only the decoder half.
- It can only read from left-to-right, meaning it is strictly blind to future words.
- **Next-Word Prediction at Scale:** Proved that if you make a decoder model large enough, it becomes an incredible creative writer without needing special instructions.
- Performs best on text generation, and chatbot-based tasks.



BART: Bidirectional and Auto-Regressive Transformer

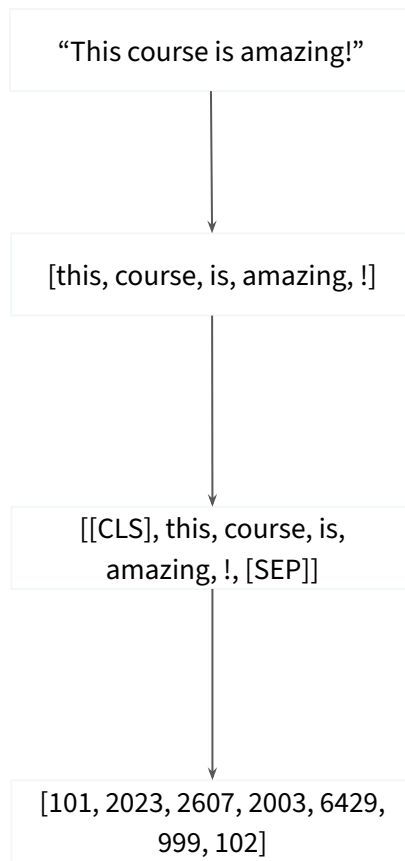
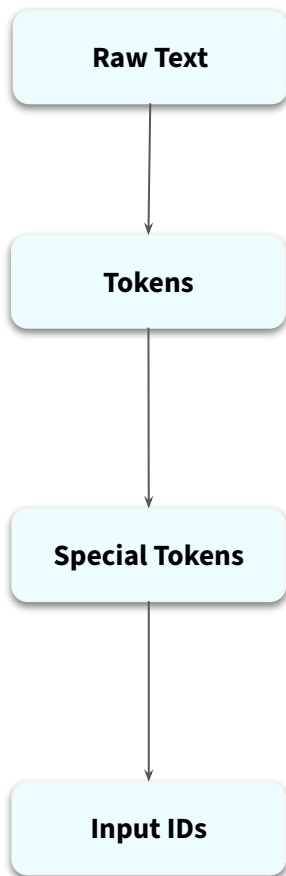
Encoder-Decoder

- Combines BERT's ability to read in any direction with GPT's ability to generate text.
- Trained by intentionally messing up text (shuffling sentences, deleting words) and forcing the model to rebuild the original text perfectly.
- Performs best on fill-mask, text summarization, translation, sentence rephrasing.

The AutoTokenizer Family

Hugging Face Basics







BertTokenizer

BERT

GPT2Tokenizer

GPT2

T5Tokenizer

T5



AutoTokenizer.from_pretrained("model-name")

Tokenization Strategies

Hugging Face Basics



Word-based Tokenizers

This course is amazing!

Word-based Tokenizer

[this, course, is, amazing, !]

[2023, 2607, 2003, 6429, 999]



Out of Vocabulary Tokens

[this, course, is, gonna, be, amazing, !]



[2023, 2607, 2003, [UNK], 1083, 6429, 999]

[this, course, is, gonna, be, sublime, !]



[2023, 2607, 2003, **[UNK]**, 1083, **[UNK]**, 999]

Character-based Tokenizers

a → 1

b → 2

c → 3

d → 4

e → 5

...

1 → 27

2 → 28

...

! → 37

...

à → 256

This course is amazing!

Character-based Tokenizer

['T', 'h', 'i', 's', ' ', 'c', 'o', 'u', 'r', 's', 'e', ' ', 'i', 's', ' ', 'a', 'm', 'a', 'z', 'i', 'n', 'g', '!']

[20, 8, 9, 19, 3, 15, 21, 18, 19, 5, 9, 19, 1, 13, 1, 26, 9, 14, 7, 37]



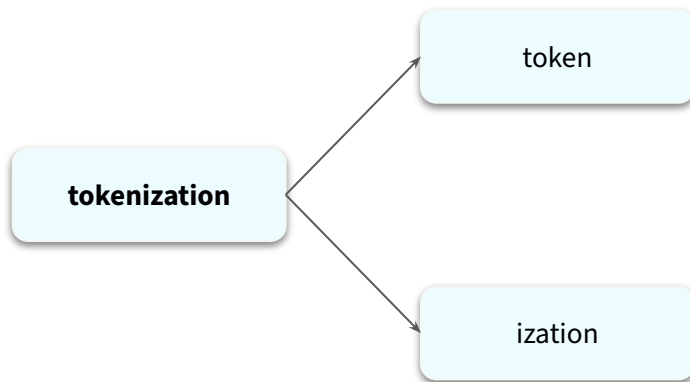
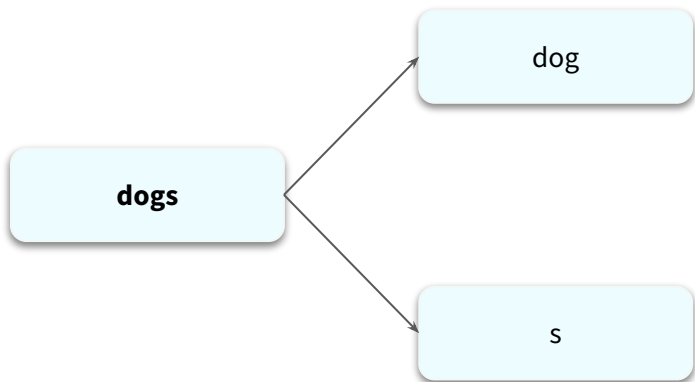
Subword-based Tokenizers

Word-based Tokenizers

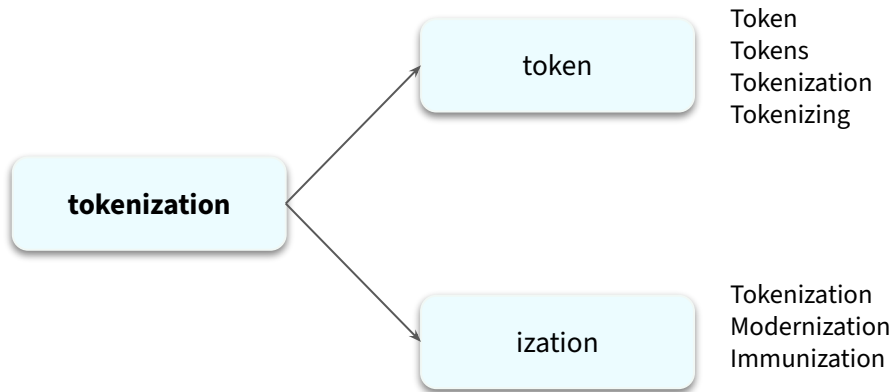
Large Vocabularies
Large Quantities of OOV words
Loss of meaning across similar words (dog vs dogs)

Character-based Tokenizers

Long Input ID Sequences
Less meaningful tokens



Subword-based Tokenizers



Most popular strategies:

- **Word Piece:** BERT, DistillBERT
- **Unigram:** XLNet, ALBERT
- **Byte-Pair Encoding:** GPT2, ROBERTa

The AutoModel Family

Hugging Face Basics

Model





AutoModel.from_pretrained("model-name")



AutoModel Task Variants

Class Name	Typical Tasks	Typical Output
AutoModelForSequenceClassification	Sentiment Analysis, Text Classification, NLI	Logits per class
AutoModelForQuestionAnswering	Extractive QA (e.g., SQuAD)	Start and End Logits for Answer Span
AutoModelForCausalLM	Text Generation, Chatbots, Code Generation	Sequence of next-token probabilities
AutoModelForSeq2SeqLM	Summarization, Translation, Text-to-Text	Generated target sequence

Hugging Face Datasets

Hugging Face Basics



pip install datasets

```
from datasets import load_dataset  
dataset = load_dataset("data-name")
```



Standard Padding (padding to global max)

Total Padding Tokens: 32

Batch 1								
the	sun	is	shining	today	PAD	PAD	PAD	PAD
a	quick	brown	fox	jumps	over	the	lazy	dog
she	drinks	hot	tea	PAD	PAD	PAD	PAD	PAD
Batch 2								
they	walked	to	the	big	park	PAD	PAD	PAD
i	forgot	my	keys	PAD	PAD	PAD	PAD	PAD
the	new	movie	is	very	bad	PAD	PAD	PAD
Batch 3								
learning	new	concepts	is	fun	PAD	PAD	PAD	PAD
i	love	coding	PAD	PAD	PAD	PAD	PAD	PAD
my	friend	plays	the	guitar	PAD	PAD	PAD	PAD

Dynamic Padding (padding to batch max)

Total Padding Tokens: 13

Batch 1								
the	sun	is	shining	today	PAD	PAD	PAD	PAD
a	quick	brown	fox	jumps	over	the	lazy	dog
she	drinks	hot	tea	PAD	PAD	PAD	PAD	PAD
Batch 2								
they	walked	to	the	big	park			
i	forgot	my	keys	PAD	PAD			
the	new	movie	is	very	bad			
Batch 3								
learning	new	concepts	is	fun				
i	love	coding	PAD	PAD				
my	friend	plays	the	guitar				